Curatte — Agentic Commerce Scraper

Technical Overview: Architecture, Bot-Wall Evasion, Cost Model & V2 Roadmap

Setup & Execution

1. Install: `pip install -r requirements.txt && playwright install chromium`

2. Configure: Copy `.env.example` → `.env` and set `OPENAI_API_KEY`, `OPENAI_MODEL` (default gpt-4o-mini), and `PROXY_URL` (residential proxy, e.g. Bright Data/Oxylabs). Secrets are never committed; `.gitignore` protects `.env`.

3. Run: `python main.py --brand adidas` (add `--verbose` for agentic decision logs). Output: `output/{brand}_offers.json` plus performance metrics.

1. Architecture

An **Agentic Pipeline** built on **LangGraph**, using Python `asyncio` for concurrency and Playwright for headless browser execution. Instead of brittle hardcoded CSS selectors, LLMs dynamically generate extraction strategies. A state machine orchestrates four specialized agents:

- **Navigator** — bootstraps the Playwright context, injects residential proxies, throttles bandwidth.
- **Site Analyzer** — takes pruned DOM, uses Pydantic structured outputs to propose a routing plan (search bars, cookie banners).
- **Searcher** — executes the plan, bypasses cookie banners, lands on the brand's coupon page.
- **Extractor** — a ReAct loop: observes the page, proposes a click, validates the resulting DOM, extracts structured Offer objects (code, title, type, expiry).

2–5. Core Engineering Solutions

Bot Walls (Cloudflare/Datadome)	Playwright stealth args (`--disable-blink-features=AutomationControlled`) plus rotating Residential Proxies injected via `PROXY_URL` to bypass enterprise protections on sites like cupones.es and wired.com.
Code Reveal & Pop-ups	Extractor's ReAct loop checks `context.pages` after every click; any rogue affiliate tab is instantly killed, the original tab refocused, and the revealed modal DOM fed back to the LLM. A `clicked_selectors` memory array prevents infinite loops.
Bandwidth Cost	Navigator uses `route.intercept` to abort image/media/font/stylesheet requests, stripping pages to pure HTML — cutting bandwidth >90% and maximizing proxy-gigabyte ROI.
Prompt Injection	Scraped HTML is treated as hostile input, sandboxed in `<untrusted_content>` delimiters; a strict system-prompt override directive makes the Agent ignore any hijack attempts embedded in the page.

6. Scale, Resilience & Cost Model

- **Concurrency:** `asyncio.Semaphore` decouples planning from execution across concurrent headless-browser batches — horizontally scalable.
- **Resilience:** environmental failures (proxy drops, Navigator timeouts) and API hiccups (SiteAnalyzer JSON errors) get exponential backoff, capped at 2 retries — protects a 10,000-brand batch without wasting proxy spend on dead domains.
- **Caching:** Site Analyzer's LLM planning is cached to disk; once a layout (e.g. promopro.co.uk) is solved, future queries skip the LLM entirely.
- **Cost:** ~$0.01 LLM tokens + ~$0.005 proxy bandwidth ≈ **$0.015/brand-site** → ~**$150** projected for 10,000 rows.

7. Determinism & DOM Ordering

LLMs are locked to `temperature=0`, but selecting the "next" candidate from a pruned HTML string can shift if DOM serialization order changes. The `clicked_selectors` guardrail rules out past decisions. For V2, classification and ranking should fully decouple: the LLM only classifies candidates ("is this a coupon button?"), while a deterministic Python script clicks valid candidates in raw top-to-bottom DOM order — keeping LLM non-determinism out of traversal routing.

8. Coverage Results & Honest Failures

- **Success:** on sites with genuine alphanumeric codes (e.g. promopro.co.uk) the Extractor parsed 20+ valid codes.
- **Affiliate traps:** sites like trustdeals.de use "Get Deal" auto-apply redirects with no real codes — correctly flagged as `dead_end` and safely aborted rather than polluting the dataset.
- **Hidden elements:** a few sites had correct selectors but Playwright threw `Element is not visible` errors — likely mobile-first menus or obfuscated overlays hiding elements from headless viewports.

9. V2 Roadmap

- **AWS Serverless:** move from local asyncio loops to Event-Driven SQS + Lambda — spin up a browser per batch, process, and die, for infinite horizontal scale.
- **LLM Model Routing:** cheap/fast models (Claude Haiku, Gemini Flash) for simple classification (Site Analyzer); reserve heavy models (GPT-4o) for complex DOM reasoning (Extractor).
- **Frontend UI:** a React/Next.js dashboard showing agents working live, backed by a stricter typed Pydantic/LangGraph state implementation.
- **Enterprise Proxies:** Bright Data Web Unlocker-style rotators to natively handle CAPTCHAs/JS challenges on restrictive domains (e.g. wired.com).
- **Multimodal Vision Agents:** feed the Extractor Playwright screenshots instead of pruned HTML so it can "read" codes visually, bypassing DOM obfuscation of class names.